

PROCESS OR MATERIALS REVIEW REQUEST FORM

Request Title: AZ-400K wet etching of Al_{0.8}Sc_{0.2}N

Requestor: Payton Broaddus

Date: 2020/11/24

Non-SNF staff, please complete the following:

Badger ID: Broaddus

Company name/PI: solgaard

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the [Stanford Chemical Storage Groups](#) to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

Material: 500nm Al_{0.8}Sc_{0.2}N on 100nm Mo on 500um Si

-Al_{0.8}Sc_{0.2}N sputtered

-Mo evaporated

-Si

Classified: Contaminated

Chemicals:

AZ-400K Developer

- 85% Water

- 15% Potassium borates

KOH active etchant

MSDS sheet: <https://cpb-us-e1.wpmucdn.com/sites.usc.edu/dist/a/194/files/2018/07/AZ400K-Developer-2a5ha25.pdf>

2. **Vendor/manufacture info:** address and phone number, website URL.

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AZ-400K: AZ Electronic Materials USA Corp. 70 Meister Avenue Somerville, NJ 08876,
Telephone No.: +1 800-515-4164

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

My device requires etching through 500nm of Al_{0.8}Sc_{0.2}N on 100nm of Mo. In order to not etch through the Mo underneath the 500nm Al_{0.8}Sc_{0.2}N, I will do a two etch process: etch 300-400nm of Al_{0.8}Sc_{0.2}N with standard Al dry etching recipes, and then I need to remove the last 100-150nm of Al_{0.8}Sc_{0.2}N with a wet etch. Wet etching of AlN has been successfully achieved using AZ-400K: <https://aip.scitation.org/doi/pdf/10.1063/1.114980>

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that if the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should be MOS grade or better.

Starting Material Stack:

500nm Al_{0.8}Sc_{0.2}N on 100nm Mo on 500um Si (100)

- 1) Headway2: Spin 4um of SPR220-3 on a 8" wafer
- 2) 110C Oven: Bake PR for 2 minutes at 110C, use a high purity Al foil on tray to hold the wafer
- 3) Scribe and dice into 4 QW
- 4) Wafersaw: cut into 60x60mm chips
- 5) Wbflexsolv: remove PR and blue tape
 - 10 min Acetone Soak
 - 10 min Methanol Soak
 - 10 min Isopropyl Soak
- 3 minutes NI gun dry
- 5a) Stresstest: check for wafer bow, get approval on Heidelberg
- 6) Yes-oven: standard surface prep
- 7) Evg spraycoat: 4um accumist coating of chip

8) Heidelberg: patterning

9) 3 hour wait time

10) Headway2:

2 minute develop in MF-24A

1 minute DI water rinse

2 minute 110C post-bake

11) ~~pt-mtl~~

~~Carrier wafer with chip held on via vacuum oil (small dip)~~

~~Std AlN etch recipe: 300nm etch (requires characterization)~~

~~Ion milling~~

12) samco

Std PR removal recipe

13) wb-res-strip

Srs-100, 20 minutes 60C, waste collected

———NEW PROCESS———

14) wb-flex-corr

AZ-400K Room temperature, time requires characterization

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

AZ-400K is a liquid developer currently in lab. The chemical is owned by Bowen Ling (who has allowed me to use some of it).

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). [Storage groups](#) A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while [storage groups](#) C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

AZ-400K stored exactly as found: in back in chemical closet.

7. **Disposal:** How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the [SNF Lab members Safety Manual](#) for the different methods of waste disposal that are available in the lab.

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Standard hazardous waste collection: all chemicals used in step 13 and 14 will be collected as described:

SRS-100 or PRS1000 must be collected in waste containers, labeled and sent out for hazardous waste disposal and must absolutely not be aspirated or poured down the rinse drain.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: